

SmartDRM: Blockchain based Digital Rights Management System

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Problem Statement

- ▶ Centralized control leading to single-point failure
- ▶ Lack of transparent ownership verification
- ▶ Manual and unreliable license enforcement
- ▶ No intelligent mechanism to detect abnormal usage patterns
- ▶ Limited auditability and accountability

Objective

- ▶ To design a decentralized DRM framework using blockchain technology
- ▶ To ensure immutable ownership and license records through smart contracts
- ▶ To provide controlled and flexible access mechanisms for digital assets
- ▶ To detect suspicious usage behavior using AI-based analysis
- ▶ To maintain an auditable and transparent activity log
- ▶ To develop a modern, user-friendly interface for asset management

Existing System

- ▶ Current DRM solutions primarily operate using centralized servers managed by content distributors or platform providers.
- ▶ Characteristics:
- ▶ Central authority controls access and licenses
- ▶ User authentication and permissions stored centrally
- ▶ DRM policies enforced at application level

Disadvantage

- ▶ Single-point failure and high vulnerability to attacks
- ▶ Limited transparency in ownership verification
- ▶ Dependency on trusted third parties
- ▶ Difficult to audit access and usage
- ▶ Poor scalability for large digital ecosystems

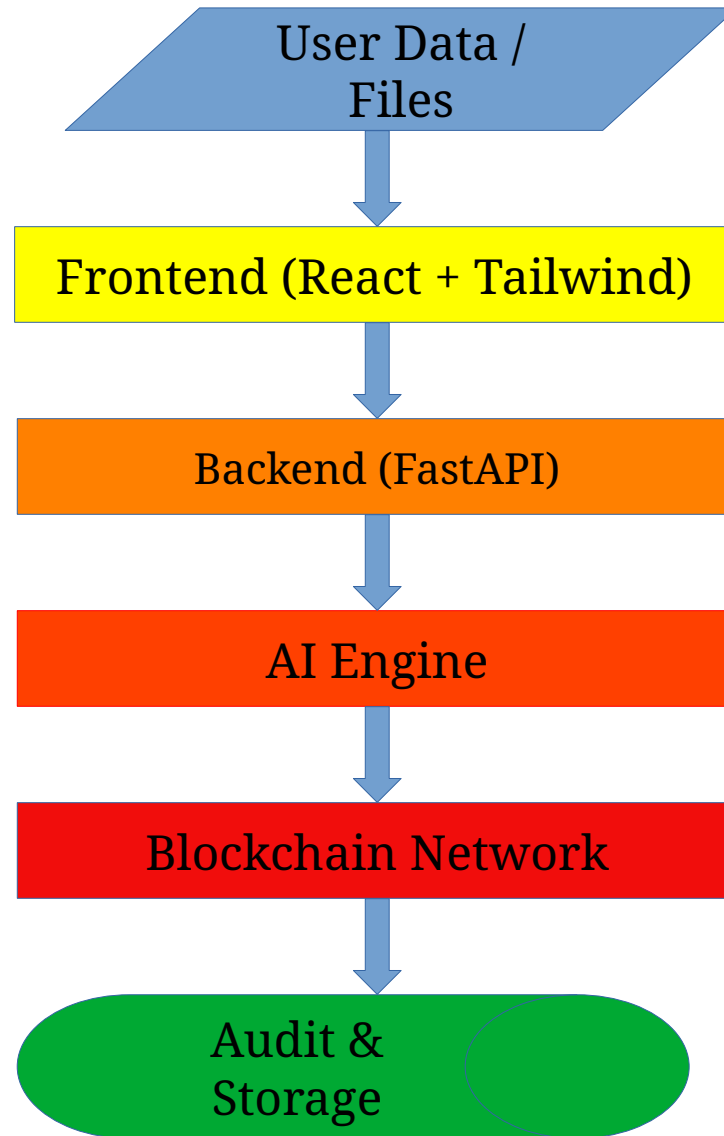
Proposed System :

- ▶ Blockchain-based asset registration and ownership proof
- ▶ Smart contract-driven license issuance and enforcement
- ▶ AI-powered behavioral risk analysis
- ▶ Secure audit logging for all critical actions
- ▶ Full-stack web interface for real-time interaction

Advantage

- ▶ Immutable and tamper-proof ownership records
- ▶ Transparent license enforcement
- ▶ Reduced piracy and unauthorized access
- ▶ Automated royalty-ready transaction tracking

SYSTEM ARCHITECTURE/FLOW DIAGRAM



Modules :

- ▶ Authentication Module
- ▶ Asset Management Module
- ▶ Blockchain & Smart Contract Module
- ▶ License & Access Control Module
- ▶ AI-Based Risk Analysis Module
- ▶ Audit Logging Module
- ▶ Frontend Dashboard Module

MODULE 1: Authentication Module

- ▶ Purpose: Secure onboarding and identity verification of all users accessing protected digital assets.
- ▶ Key Features: Email/password login, secure password hashing, and session/token-based authentication.
- ▶ Role-Based Access: Differentiates admins, content owners, and consumers with tailored permissions.
- ▶ Security Controls: Brute-force protection, validation of credentials, and secure storage of user data.
- ▶ Integration Points: Acts as the entry gate to all other modules (assets, blockchain, AI risk, and dashboard).

MODULE 2: Asset Management

- ▶ Purpose: Central hub to register, encrypt, store, and manage digital content.
- ▶ Asset Lifecycle: Upload, metadata registration, encryption, storage, and controlled distribution.
- ▶ Content Protection: Uses cryptographic techniques to prevent unauthorized duplication and sharing.
- ▶ Search & Organization: Metadata-based cataloging for quick discovery and filtering of assets.
- ▶ Integration Points: Links assets to licenses, blockchain records, and audit logs.

Module 3: Blockchain & Smart Contract

- ▶ Purpose: Provide tamper-evident, verifiable records for licenses and asset transactions.
- ▶ Smart Contracts: Encode licensing terms (duration, usage limits, regions) as executable logic.
- ▶ Immutable Ledger: Stores key events like license issuance, revocation, and critical asset operations.
- ▶ Transparency & Trust: Enables independent verification of rights and reduces disputes.
- ▶ Integration Points: Consumed by license/access control and audit modules for verification.

Conclusion

- ▶ SmartDRM-X demonstrates how blockchain and artificial intelligence can be effectively integrated to build a secure, transparent, and intelligent Digital Rights Management system. By decentralizing ownership records, automating license enforcement, and introducing behavioral analysis, the system significantly improves trust and security in digital content distribution.
- ▶ The proposed framework is scalable, extensible, and suitable for applications in education, digital media platforms, enterprise document management, and software licensing.
- ▶ Future enhancements may include integration with decentralized storage systems and advanced machine learning models for predictive piracy detection.



THANK YOU !